

Chapter 5

Infinite series

1. Infinite series

$$\begin{array}{cccccc} x_1 & x_2 & x_3 & \cdots & x_n & \cdots \\ \frac{1}{2} & \left(\frac{1}{2}\right)^2 & \left(\frac{1}{2}\right)^3 & \cdots & \left(\frac{1}{2}\right)^n & \cdots \end{array}$$

The series is $\sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n$

Definition

$$S_n = \sum_{n=0}^{\infty} a_n$$

$$\lim_{N \rightarrow \infty} S_N = C \quad \text{finite}$$

We call this series converges, otherwise this series diverges.

$$\begin{aligned} \sum_{n=0}^N a^n &= \frac{1 - a^{N+1}}{1 - a} \quad \sum_{n=n_1}^{n_2} a^n = \frac{a^{n_1} - a^{n_2+1}}{1 - a} \\ a &= \frac{1}{2}, \quad \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1 - 0}{1 - \frac{1}{2}} = 2. \end{aligned}$$

A infinite series $\sum_{n=0}^{\infty} u_n$, if $u_0 + u_1 + u_2 + \cdots + u_n + \cdots = C$, we can get

$$\lim_{n \rightarrow \infty} u_n = 0$$

Proof

$$S_N = \sum_{n=0}^N u_n, \quad \lim_{N \rightarrow \infty} S_N = C$$

$$T_N = \sum_{n=0}^{N+1} u_n, \quad \lim_{N \rightarrow \infty} T_N = C$$

$$T_N - S_N = u_{N+1}, \quad \lim_{N \rightarrow \infty} T_N - S_N = 0$$

Here are some ways to determine the convergence of infinite series

1. Comparison test (converge)

If $\{b_n\}$ converges and $b_n > 0$, $\sum_{n=0}^{\infty} b_n = C$, and $|a_n| \leq |b_n|$, then $\{a_n\}$ converges

Example 1

$$\left\{ \frac{1}{n!} \right\} \quad (n! = 1 \times 2 \times 3 \times \cdots \times n)$$

according to

$$\frac{1}{1 \times 2 \times 3 \times 4} < \frac{1}{2 \times 2 \times 2 \times 2}$$

We try to proof

$$\frac{1}{n!} < \frac{1}{2^n}$$

for some n

Example 2

$$\begin{aligned} & \left\{ \frac{1}{n(n-1)} \right\} \quad n = 2, 3, \dots \\ & \frac{1}{1 \times 2}, \frac{1}{2 \times 3}, \frac{1}{3 \times 4} \dots \\ = & \frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots \\ = & 1 - \frac{1}{n} \\ & \lim_{n \rightarrow \infty} S_n = 1 \end{aligned}$$

Besides, $\frac{1}{n!} < \frac{1}{n(n-1)}$, so $\frac{1}{n!}$ converges definitely.

2. Comparison test (diverge)

If $\{b_n\}$ diverges and $b_n > 0$, $\sum_{n=0}^{\infty} b_n = C$, and $|a_n| > b_n$, then $\{a_n\}$ diverges

Example 3

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}$$

$$\begin{aligned} a_1 &= 1 &> 1 = b_1 \\ a_2 &= \frac{1}{2} &> \frac{1}{2} = b_2 \\ a_3 &= \frac{1}{3} + \frac{1}{4} &> \frac{1}{4} + \frac{1}{4} = \frac{1}{2} = b_3 \\ a_4 &= \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} &> \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = b_4 \\ a_5 &= \frac{1}{9} + \cdots + \frac{1}{16} &> \frac{1}{16} + \frac{1}{16} + \cdots + \frac{1}{16} = \frac{1}{2} = b_5 \end{aligned}$$

$$\begin{aligned} 1 + \sum_{n=2}^{\infty} \frac{1}{2} &= \infty \\ \Rightarrow \sum a_n &= \infty \\ \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n} &= \infty \end{aligned}$$

Example 4

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < \sum_{n=1}^{\infty} \frac{1}{n(n-1)}$$

So, it converges.

3. Ratio Test

Given $\{u_n\}$, if $\left| \frac{u_{n+1}}{u_n} \right| < \rho < 1$, ρ does not depend on n . Then, $\{u_n\}$ converges.

Example 5

$$\left\{ \frac{1}{n!} \right\}$$

$$\left| \frac{u_{n+1}}{u_n} \right| = \frac{1}{n+1} < \frac{1}{2} < 1 \quad (n > 1)$$

So, $\left\{ \frac{1}{n!} \right\}$ converges.

4. Cauchy Test

$$\sqrt[n]{|u_n|} < a \quad |a| < 1$$

Given a sequence $\{u_n\}$, if $\lim_{n \rightarrow \infty} |u_n|^{\frac{1}{n}} = c < 1$, then $\{u_n\}$ converges

2. zero transform

The infinite series of z

Given a sequence of numbers

$$\cdots, 0, 0, 0, x_1, x_2, x_3, \cdots$$

$$\cdots, 0, 0, 0, 5, -3, 2, \cdots$$

x_n is causal if $x_n = 0, n < 0$

If x_n is infinite, there are infinite non-zero terms. If x_n is finite, there are finite non-zero terms.

Example 1

$$\sum_{n=0}^{\infty} x_n \cdot Z^{-n} = x(z)$$

$$x_n = 0.5, 3, -2, 1$$

$$x(z) = 0.5 + 3z^{-1} - 2z^{-2} + z^{-3}$$

Example 2

Haar wavelet

$$H_0 \quad \left\{ \frac{1}{2}, \frac{1}{2} \right\}$$

$$H_1 \quad \left\{ \frac{1}{2}, -\frac{1}{2} \right\}$$

We can get,

$$H_0(z) = \frac{1}{2} + \frac{1}{2}z^{-1}$$

$$H_1(z) = \frac{1}{2} - \frac{1}{2}z^{-1}$$

Example:

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

$$x[n] = a^n u[n]$$

$$x[n] = \left(\frac{1}{2}\right)^n u[n]$$

The sequence is

$$1 \quad 1/2 \quad 1/4 \quad 1/8 \quad 1/16 \quad \dots$$

If $\left|\frac{1}{2}z^{-1}\right| < 1$, then

$$\begin{aligned} x(z) &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n \cdot z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n \\ &= \frac{1 - \left(\frac{1}{2}z^{-1}\right)^{\infty}}{1 - \frac{1}{2}z^{-1}} \\ &= \frac{1 - 0}{1 - \frac{1}{2}z^{-1}} \\ &= \frac{z}{z - \frac{1}{2}} \end{aligned}$$

$$\left(\frac{1}{2}\right)^n u[n] \leftrightarrow \frac{z}{z - \frac{1}{2}}, |z| > \frac{1}{2}$$

$$a^n u[n] \leftrightarrow \frac{z}{z - a}, |z| > |a|$$

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

$$\delta[n] \leftrightarrow 1$$

$$u[n] \leftrightarrow \frac{z}{z - 1}$$

$$\delta[n - 5] \leftrightarrow z^{-4}$$

$$\delta[n - 6] \leftrightarrow z^{-5}$$

If $a = e^{in\theta}$,

$$e^{in\theta}u[n] \leftrightarrow \frac{z}{z - e^{i\theta}}$$

There are two sequences.

$$\begin{array}{ccccccc} x_1 & x_2 & x_3 & \cdots & x_n & \cdots & \leftrightarrow x(z) \\ y_1 & y_2 & y_3 & \cdots & y_n & \cdots & \leftrightarrow y(z) \end{array}$$

Suppose we have a new sequence

$$x_1 + y_1 \quad x_2 + y_2 \quad x_3 + y_3 \quad \cdots \quad x_n + y_n \quad \cdots \quad \leftrightarrow x(z) + y(z)$$

$$\sum (x_n + y_n)z^{-n} = \sum x_n z^{-n} + \sum y_n z^{-n}$$

$$e^{-in\theta}u[n] \leftrightarrow \frac{z}{z - e^{-i\theta}}$$

$$[e^{in\theta} + e^{-in\theta}]u[n] \leftrightarrow \frac{z}{z - e^{i\theta}} + \frac{z}{z - e^{-i\theta}}$$

$$\left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n] \leftrightarrow \frac{z}{z - \frac{1}{2}} + \frac{z}{z - \frac{1}{3}}$$

$$2, 0.833, \dots \leftrightarrow \frac{2z - 5/6}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{3}\right)} \cdot z$$

Partial fraction

$$\begin{aligned} \frac{z}{(z+1)(z-1)} &= z \left[\frac{A}{z+1} + \frac{B}{z-1} \right] \\ &= -0.5 \frac{z}{z+1} + 0.5 \frac{z}{z-1} \end{aligned}$$

$$x[n] = -0.5(-1)^n u[n] + 0.5 \cdot 1^n \cdot u[n]$$

Example:

$$\frac{z}{z^2 - 0.8z + 0.45} = \frac{z}{(z - 0.3)(z - 0.5)} = -5 \frac{z}{z - 0.3} + 5 \frac{z}{z - 0.5}$$

Rational function of z

$$\begin{aligned}f(z) &= \frac{Q(z)}{P(z)} \\&= \frac{R_1}{z - P_1} + \frac{R_2}{z - P_2} + \cdots + \frac{R_N}{z - P_N}\end{aligned}$$

$$P(z) = (z - p_1)(z - p_2) \cdots (z - p_N)$$

$$P_1 \neq P_j$$

$$\begin{aligned}P(z) &= z^2 - 0.8z + 0.15 \\&= (z - 0.3)(z - 0.5)\end{aligned}$$

Example

$$\begin{aligned}f(z) &= \frac{z^2}{(z - 0.3)(z - 0.5)} \\&= z \frac{z}{(z - 0.3)(z - 0.5)} \\&= z \left[\frac{k_1}{z - 0.3} + \frac{k_2}{z - 0.5} \right]\end{aligned}$$

$$k_1 = \frac{z}{z - 0.5} \Big|_{z=0.3} = \frac{0.3}{0.3 - 0.5} = -\frac{3}{2}$$

$$k_2 = \frac{z}{z - 0.3} \Big|_{z=0.5} = \frac{0.5}{0.5 - 0.3} = \frac{5}{2}$$

Example $\frac{z+1}{(z-0.3)(z-0.2)(z-0.3)}$

$$\frac{z+1}{(z-0.5)(z-0.2)} \Big|_{z=0.3} = \frac{1.3}{-0.2 \times 0.1} = -65$$

$$\frac{z+1}{(z-0.5)(z-0.3)} \Big|_{z=0.2} = \frac{1.2}{0.3 \times 0.1} = 40$$

$$\frac{z+1}{(z-0.3)(z-0.3)} \Big|_{z=0.5} = \frac{1.5}{0.3 \times 0.2} = 25$$

$$\frac{z+1}{(z-0.5)(z-0.3)(z-0.3)} = \frac{25}{z-0.5} + \frac{40}{z-0.2} - \frac{65}{z-0.3}$$

Repeated roots

$$\frac{z}{(z-0.3)^2(z-0.5)} = \frac{k_1}{z-0.5} + \frac{k_2}{z-0.3} + \frac{k_3}{(z-0.3)^2}$$

$$\frac{z}{(z-0.3)^3(z-0.5)} = \frac{k_1}{z-0.5} + \frac{k_2}{z-0.3} + \frac{k_2}{(z-0.3)^2} + \frac{k_4}{(z-0.3)^3}$$

For $\frac{z}{(z-0.3)^2(z-0.5)}$

$$k_1 = \frac{0.5}{(0.5 - 0.3)^2} = \frac{0.5}{(0.2)^2} = 12.5$$

$$k_3 = \frac{0.3}{0.3 - 0.5} = -\frac{3}{2}$$

Let $z = 0$

$$0 = \frac{12.5}{-0.5} + \frac{k_2}{-0.3} + \frac{-1.5}{0.09} \Rightarrow k_2$$

We can solve these kinds of problems with a function of MATLAB , *residue()*

Example $\frac{z+1}{(z-0.9)(z-0.5)(z-0.2)}$, find its inverse Z -transform

$$\begin{aligned} \frac{z+1}{(z-0.9)(z-0.5)(z-0.2)} &= z \frac{z+1}{z(z-0.3)(z-0.5)(z-0.2)} \\ &= k_1 \delta[n] + k_2 (0.3)^n u[n] + k_3 (0.5)^n u[n] + k_4 (0.2)^n u[n] \end{aligned}$$

$$\frac{z}{z-a} = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$\left(\frac{z}{z-a} \right)' = \frac{1}{z-a} + z \frac{1}{(z-a)^2} (-1) = \frac{-a}{(z-a)^2}$$

$$\left(\sum_{n=0}^{\infty} a^n z^{-n} \right)' = \sum_{n=0}^{\infty} a^n z^{-(n+1)} \cdot (-n) = \sum_{n=0}^{\infty} b_n z^{-n}$$

Then we can get,

$$\frac{1}{(z-1)^2} = \sum_{n=1}^{\infty} (n-1) z^{-n}$$